School of Computing

Assignment 2 – Analysing the web browsing process

CSC1010 – Networks and Internet

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Objective

This assignment is designed to help students understand the web browsing process by analysing network topology and performing hands-on simulations in Packet Tracer.

Part 1: Observing the Topology

Open the file “Assignment2.pkt”. The topology consists of:

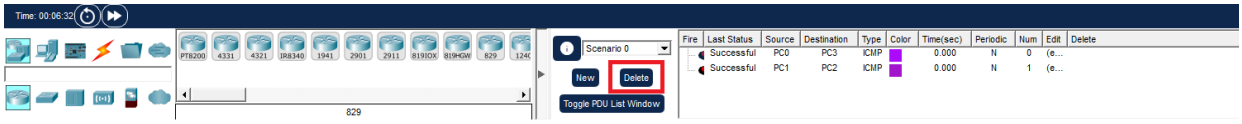
- 1 computers (PC)
- 1 switch
- 2 routers (Router 1 and Router 2)
- 2 servers (DNS server and Web Server)

Answer the following questions about the topology:

Q1. How many networks are there in the topology?

Part 2: Analysing the web browsing process

- Click on the Closed Envelope icon at the top menu to create a message. Then, click on PC and DNS Server
- Click again on the Closed Envelope icon and click on PC and Web Server.
- Click on the Delete button at the bottom centre of the Cisco Packet Tracer.



- Switch to Simulation Mode by clicking on the "Simulation Mode" icon at the bottom-right corner of Packet Tracer.
- Click on PC and select the Desktop tab. Click then on the Web Browser icon.
- Type in www.cisco.com
- Click on the Go button

Q2. What is the type of message generated?

Q3. At what layer of the OSI model the message was created?

Q4. What transport layer protocol is used to encapsulate the message?

Q5. What is the source port?

Q6. What is the destination port?

Q7. What is the source IP address? Give just the IP address and nothing else.

Q8. What is the destination IP address? Give just the IP address and nothing else.

Q9. To which device does the destination MAC address belong to?

- Click on the forward button on the simulation model panel. The message will be transmitted to the switch
- Click again on the forward button. The message will then be transmitted to Router 1.
- Click again on the forward button. The message will be transmitted to the DNS server.

Q10. What is the IP address mapped to the domain name www.cisco.com? Give just the IP address and nothing else.

- Click on the forward button. The response message will be transmitted to Router 1. Keep clicking on the forward button until the message is delivered to PC.
- A new message is generated at PC (green envelop).

Q11. To which protocol does the message belong? Only give the protocol name abbreviated.

Q12. What is the name of the TCP process that is triggered? Give just the name of the process and nothing else

- Click on the message and select the Outbound PDU Details tab. Look at the TCP header and answer the following questions:

Q13. What is the source port number?

Q14. What is the destination port number?

Q15. What is the sequence number?

Q16. Which of the following flags are set to one (URG, ACK, PSH, RST, SYN, FIN)?

- Keep clicking on the forward button until the message reaches the Web Server. Click on the message and click on the Outbound PDU Details tab. Look at the TCP header and answer the following questions:

Q17. Which of the following flags are set to one (URG, ACK, PSH, RST, SYN, FIN)?

Q18. What is the Acknowledgement Number?

- Keep clicking on the forward button until the message reaches PC. Click on the message and click on the Outbound PDU Details tab. Look at the TCP header and answer the following questions:

Q19. Which of the following flags are set to one (URG, ACK, PSH, RST, SYN, FIN)?

Q20. What is the sequence number?

Q21. What is the acknowledgment number?

- Click on the newly generated message (purple envelop).

Q22. What type of message is it?

- Click on the Outbound PDU Details tab. Look at the TCP header and answer the following question:

Q23. What is the destination port number?

Q24. Which of the following flags are set to one (URG, ACK, PSH, RST, SYN, FIN)?

- Keep clicking on the forward button until both messages are received by the Web Server.

- Click on the message and then click on Layer 7 of the Out Layers.

Q25. What is the type of the message generated?

- Keep clicking on the forward button until the message reaches PC.

- Click on PC > Desktop > Web Browser. You will see now the requested web paged displayed.

- If you close the PC window. You will see that another message was generated (green envelop). Click on it and click on the Outbound PDU Details tab. Look at the TCP header and answer the following questions:

Q26. Which of the following flags are set to one (URG, ACK, PSH, RST, SYN, FIN)?

Q27. What does the ACK bit indicate here?

Q28. What is the name of the TCP process that is triggered?

- Keep clicking on the forward button until the message reaches the Web Server. Click on it and click on the Outbound PDU Details tab. Look at the TCP header and answer the following question:

Q29. Which of the following flags are set to one (URG, ACK, PSH, RST, SYN, FIN)?

- Keep clicking on the forward button again until the message reaches PC. Click on it and click on the Outbound PDU Details tab. Look at the TCP header and answer the following question:

Q30. What does the ACK bit indicate here?

- Keep clicking on the forward button until the message reaches the Web Server.

Guidelines

- Use **Assignment 2 – Answer Sheet** on LOOP to submit your answers.
- Be concise and precise; avoid lengthy responses. Use numerical values to answer “how many” questions.
- The **deadline is 22nd February 2026, 23:59.**
- This assignment contributes **10%** to your final mark.